European Journal of Remote Sensing

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/tejr20

Wind retrieval of Fabry-Perot interferometer in the middle and upper atmosphere using wavelength depth method based on satellite-based simulation and ground-based measurement

Houmao Wang , Huanhuan Yan , Liping Fu , Xiaoxin Zhang & Weiguo Zong

To cite this article: Houmao Wang , Huanhuan Yan , Liping Fu , Xiaoxin Zhang & Weiguo Zong (2020) Wind retrieval of Fabry-Perot interferometer in the middle and upper atmosphere using wavelength depth method based on satellite-based simulation and ground-based measurement, European Journal of Remote Sensing, 53:1, 145-155, DOI: 10.1080/22797254.2020.1752810

To link to this article: https://doi.org/10.1080/22797254.2020.1752810

© 2020 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

Published online: 18 Apr 2020.

Submit your article to this journal

Article views: 642

View related articles

View Crossmark data

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tejr20

EUROPEAN JOURNAL OF REMOTE SENSING
2020, VOL. 53, NO. 1, 145–155
https://doi.org/10.1080/22797254.2020.1752810

Taylor & Francis
Taylor & Francis Group

OPEN ACCESS Check for updates

Wind retrieval of Fabry-Perot interferometer in the middle and upper atmosphere using wavelength depth method based on satellite-based simulation and ground-based measurement

Houmao Wang[a,b,c], Huanhuan Yan[d], Liping Fu[a,b,c], Xiaoxin Zhang[d] and Weiguo Zong[d]

[a]Laboratory of Space Environment Exploration, National Space Science Center, Chinese Academy of Sciences, Beijing, China; [b]Beijing Key Laboratory of Environment Exploration, Beijing, China; [c]Key Laboratory of Environmental Space Situation Awareness Technology, Chinese Academy of Sciences, Beijing, China; [d]National Satellite Meteorological Center, China Meteorological Administration, Beijing, China

ABSTRACT

The satellite-based observations of the Fabry-Perot Interferometer interference ring in the middle and upper atmosphere (80–300 km) are simulated, which include the 557.7 nm, 630.0 nm, and O_2 (0–0) airglows. Based on the simulation, the central wavelength of the airglow is determined to calculate the wind. First, based on the relationship model between the wavelength and peak position of the interferometer fringe, the peak position is adjusted by changing the wavelength until it is equal to the measured peak position. Then, the nominal wind is calculated by combining the obtained wavelength with the assumed wavelength of zero wind speed. Finally, the final wind is obtained by the zero reference correction. To validate the method, the scheme is used for wind retrieval, and the results are compared with the actual winds, which are the input parameters in the simulation. The average errors of the 557.7 nm and 630.0 nm dayglow are 3.18 m/s and 3.29 m/s, respectively. The average errors of the nightglow are slightly larger at 3.79 m/s (557.7 nm) and 9.08 m/s (630.0 nm), owing to their weak radiation. The average error of O_2 (0–0) is 1.41 m/s, while it is 5.88 m/s using the traditional method. These results show that this method is applicable to wind retrieval.

ARTICLE HISTORY
Received 6 September 2019
Revised 22 March 2020
Accepted 3 April 2020

KEYWORDS
Fabry-Perot interferometer; wind retrieval; airglow; satellite-based simulation; ground-based measurement; wavelength depth method

Introduction

The neutral wind is one of the most basic and important parameters of the middle and upper atmosphere. It not only plays an important role in the energy transmission and dynamics of the atmosphere (Larsen & Meriwether, 2012) but also has a great impact on the safety and orbits of spacecraft, along with the atmospheric density and temperature (Cai et al., 2007; Chen et al., 2013). Therefore, wind measurements can be used for the study of the coupling of atmospheric dynamics and photochemistry. At present, the main passive optical instruments used for the neutral wind measurement are the Fabry-Perot interferometer (FPI), Michelson interferometer, and asymmetric spatial heterodyne interferometer. Among them, owing to its high sensitivity and large light transmittance, the FPI is widely used for the neutral wind measurements of the middle and upper atmosphere, which comprise both ground- and satellite-based measurements.

Instrument

The FPI has been used to detect the winds of the middle and upper atmosphere since the 1960 s. In 1982, the FPI onboard the DE-2 satellite (DE-FPI) was first used to obtain wind measurements (Killeen & Hays, 1984; Killeen et al., 1983). In 1991, the FPI on the Upper Atmosphere Research Satellite measured the wind, temperature, and volume emission rate by observing the airglow radiation at 50–120 km (Abreu et al., 1991; Hays et al., 1993; Skinner et al., 1987). In 2001, the FPI on the Thermosphere Ionosphere Mesosphere Energetics and Dynamics satellite (TIDI) measured the wind and temperature at 60–300 km using a single etalon and a circle-to-line system, with a minimum wind measurement error of 3 m/s (Killeen et al., 2006). To date, nighttime observations using the ground-based FPI have been carried out for several decades to obtain wind measurements (Peterson et al., 1966; Armstrong, 1969; Hernandez & Roble, 1995; Shiokawa et al., 2001, 2012; Makela et al., 2009□Kosch et al., 2010; Wu et al., 2004; Wang et al., 2016). The precision of these measurements is generally within 10 m/s, depending on the airglow intensity, instrument design, and retrieval method. Furthermore, the FPI is feasible for daytime measurements using a triple-etalon system (Gerrard & Meriwether, 2011; Wu et al., 2012) to avoid the strong Rayleigh scattering in the lower atmosphere.

Retrieval algorithm

Algorithms used to retrieve winds have been studied since the 1960 s. Hernandez (1966), 1979, Henderson

CONTACT Houmao Wang hmwang@nssc.ac.cn No.1 Nanertiao, Zhongguancun, Haidian District, Beijing, China

© 2020 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.

(1986) derived an analytical model for the FPI. They refined the defect broadening function and simulated the instrument transfer function of the FPI. Hays and Roble (1971) used this transfer function to propose a non-linear least-squares method to retrieve the atmospheric wind and temperature. The transfer function was based on the convolution of the Airy function and spherical defect function, plane irregularity defect function, and diaphragm-broadening function. However, an accurate instrument function cannot be obtained without considering the influence of the optical distortion and asymmetry. In order to solve the problem, Killeen (1983) and Killeen and Hays (1984) proposed using the complete Fourier series to represent all of the broadening functions, in which almost all defects of the optical systems are considered, greatly improving the retrieval accuracy. This method has been used in many FPI measurements to obtain the wind and temperature simultaneously.

Shiokawa et al. (2001, 2003) obtained the peak radius of the interference ring and calculated the wind speed based on the direct relationship between the peak radius and atmospheric wind speed. In this method, the relationship between the wind and interference ring radius was derived, avoiding the need for accurate FPI instrument parameters.

Harding et al. (2014) proposed a method to construct the FPI forward model based on the simpler Airy function, quadratic equation of intensity, and point-spread function. The forward model was used to retrieve the wind and temperature simultaneously based on the Levenberg-Marquardt fitting algorithm.

Contrary to the international effort regarding the FPI algorithm, studies in China did not take place until after 2000. Since then, studies have been carried out by the National Space Science Center (Wang & Wang, 2015; Wang et al., 2013, 2014, 2014, 2017). Wang et al. (2013) determined the center and radius of the incomplete interference ring simultaneously, and Wang et al. (2017) improved the algorithm to obtain a much more accurate wind speed.

Though the FPI can also be used for temperature observations, the method described in this paper only applies to wind retrieval, for two reasons. One is that the temperature is retrieved from the molecular airglow observations (< 150 km) using an independent algorithm. The other is that the accuracy of the temperature retrieval has not yet been validated or confirmed as precisely as that of wind retrieval. Therefore, the described method is only applicable to wind retrieval.

In this paper, Section 2 describes the mechanism of the forward model used for the airglow simulation and wind retrieval and also describes the wind retrieval method. Section 3 provides the details of the satellite-based simulation of the O_2 (0–0), red line 630.0 nm, and green line 557.7 nm airglows. Section 4 analyzes and concludes the discussion of the retrieval results of the satellite-based simulation. Section 5 displays the results from the ground-based FPI interferogram observations for further validation of the method. In Section 6, a brief summary is given.

Methodology

Mechanism of forward model

The core component of the FPI is a parallel glass plate with a reflective coating, i.e., an etalon. In the ideal case, its function (i.e., Airy function) is (Hernandez, 1966):

$$A = \frac{I}{\frac{1}{1+R} + \frac{4R}{(1-R)^2}\sin^2\left(\frac{2n\pi t}{\lambda}\cos\theta_i\right)} \tag{1}$$

where I is the intensity, R is the reflectance of the etalon, n is the atmospheric refractive index, t is the etalon gap, λ is the wavelength, and θ_i is the angle with the optical axis. Equation (1) shows that there is a trigonometric function relationship between λ and θ_i, where θ_i is the $\tan^{-1}$ of the ratio of the ring radius to the focal length. When the other parameters remain constant, a change in the central wavelength causes the radius of the interference ring to change. Therefore, the radius of the interference ring can be obtained once the central wavelength is determined. In addition, the point-spread function is introduced to better describe the actual function of the FPI optical system (Harding et al., 2014). Hence, Equation (1) becomes:

$$Ai = \int_0^{r_{max}} b(s,r)A(s,r)ds \tag{2}$$

where A_i is the improved Airy function, $b(s,r)$ is the point-spread function, r is the radius of the interference ring, r_{max} is the maximum of the radius, and s is a dummy variable in the same coordinate system as r. The point-spread function just widens the fringes but does not influence the peak position of the fringes. Equation (2) is convoluted with the airglow spectrum function:

$$S(r) = \int_{-\infty}^{+\infty} Ai(r,\lambda)Y(\lambda)d\lambda + B \tag{3}$$

where B is the offset value of the CCD (Charged Coupled Device), and Y is a function of the airglow intensity, background radiation, central wavelength, and spectral width of the airglow radiation. The central wavelength and spectral width are related to the wind speed and temperature, respectively. Therefore, the airglow radiation function with the change in wind speed and temperature information can be expressed as follows:

$$Y(\lambda) = Y_{bg} + Y_{line}\exp\left(-\frac{1}{2}\left(\frac{\lambda-\lambda_c}{\Delta\lambda}\right)^2\right) \tag{4}$$

where Y_{bg} is the background radiation, Y_{line} is the intensity of airglow radiation, λ_C is the central wavelength including wind speed, and $\Delta\lambda$ is the width of the Doppler broadening.

$$\lambda_c = \lambda_0\left(1 + \frac{v}{c}\right) \tag{5}$$

$$\Delta\lambda = \frac{\lambda_0}{c}\sqrt{\frac{kT}{m}}$$

where λ_0 is the initial central wavelength, v is the wind speed, c is the light speed, k is the Boltzmann constant, T is the temperature, and m is the molar mass.

Retrieval method

The forward model of the FPI is constructed based on Equation (3). The instrument parameters are determined by fitting Equation (2) to the laser data using the model (Figure 1), which includes the etalon gap, etalon reflectivity, and point-spread function.

Owing to the differences in the wavelengths between the laser calibration and airglows, it is necessary to re-convolute the instrument function with the airglow spectrum function to obtain the forward model of the FPI airglow interference. The retrieval steps are as follows:

(1) The airglow spectrum function is obtained according to Equation (4) with an initial central wavelength. The initial wavelength is assumed to be known and set near the actual value. The difference between them will not lead to errors because of the zero wind reference correction described below.

(2) Unlike the traditional methods, the airglow width (corresponding to temperature) is not considered in our method. Instead, we use a no-width spectrum to convolve with the instrument function. Then, the one-dimensional curve ($S(r)$) of the FPI airglow interference is simulated.

(3) The Gaussian function for fringe fitting is $y = a_1 e^{-\left(\frac{x-a_2}{a_3}\right)^2}$, where a_1, a_2, and a_3 are the fitting coefficients, y represents the CCD count, and x represents the CCD radius. For fitting, the Levenberg–Marquardt algorithm is used to minimize δ:

$$\delta = \sum_{i=0}^{n}\left(\frac{M(r_i) - S(r_i)}{\sigma_{r_i}}\right)^2 \tag{6}$$

where n is the number of fitting points, M is the measured signal, S is the simulated signal from Equation (3), and σ is the uncertainty of M. Based on the Gaussian fitting, the peak position (or radius) of each interference ring of the simulation (the same fitting for measurement) is obtained. Then, the difference in the interference peak position between the simulation and the actual measurement is calculated.

(4) If the difference is greater than the set threshold (0.01 pixel, which may lead to, at most, a wind error of 0.5 m/s for our FPI), the central wavelength of the airglow will be changed according to the plus-minus and magnitude of the difference. The plus-minus is used to determine the direction of the change in the central wavelength, allowing us to avoid an endless loop and find an accurate central position. The magnitude is used to

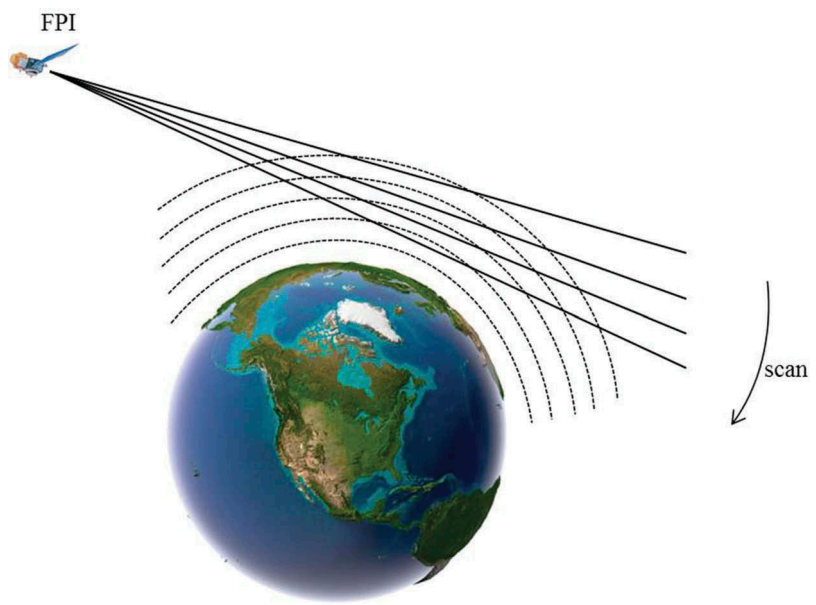

Figure 1. Limb observation of FPI.

determine the step of the central wavelength change, which is proportional to the magnitude. This can greatly improve the calculation efficiency.

(5) The steps of (2)–(4) will continue to loop sequentially until the difference in the peak position between the simulated and measured interference is within the threshold. Once the looping is completed, the drifted central wavelength is considered to be determined.

The method described above, named the wavelength depth method (WDM) in this paper, is based on the assumption that the drift of the central wavelength is caused only by wind. Thus, once the central wavelength is determined, the nominal wind speed can be obtained according to the initial central wavelength (Equation 5). Finally, the actual wind speed can be obtained by correcting the nominal wind speed with the position of zero wind speed (named zero wind reference correction in this paper), which can be determined based on several assumptions for actual satellite-based and ground-based observations (Burrage et al., 1994). However, in our study, we consider the wavelength that corresponds to that with no wind whose interference peak position is the zero wind reference position.

For other retrieval methods, the wind and temperature are calculated simultaneously by fitting interference patterns using non-linear least-squares methods (Harding et al., 2014; Killeen & Hays, 1984). These methods are also used here for wind retrieval, representing traditional methods. Their results are also shown and compared with our results in detail in Sections 4 and 5.

Satellite-based simulation

In this study, satellite limb observations of airglow are simulated for the FPI based on three wavelengths: the green line OI 557.7 nm, the red line OI 630.0 nm, and the 761.1396 nm of the O_2 (0–0) band. The observation mode of the FPI is shown in Figure 1. The green line airglow is mainly distributed above the mesopause region, the red line airglow is mainly at the lower thermosphere, and the O_2 (0–0) airglow resides at the mesosphere.

According to Equation (3), the satellite-based simulation requires instrument and airglow spectra information, which are listed in Table 1. In the simulation, the vertical field-of-view of 0.05° limits the vertical resolution of simulation to be ≤ 2.5 km. The filter transmittance function is assumed as a Gaussian function, with a peak transmittance of 0.7. Furthermore, the CCD quantum efficiency is set to be a constant value (0.3). Several mirrors are also considered; their total transmittance is ~0.41.

The line-of-sight wind speed and temperature are input values in the simulation. Figure 2 shows these input values for an altitude of 80–300 km. The wind speed is calculated using the model HWM07 (Hedin et al., 1996, 1988), which is an empirical model. Its wind data were mainly taken from rocket soundings, incoherent radar, MF radar, meteor radar, and FPI. The temperature data is taken from the International Reference Ionosphere model (IRI2012), also an empirical model, which is based on ground and space data (Bilitza, 2001). The temperature just serves to facilitate the simulation of actual airglow radiance and is not retrievable here. Furthermore, the geomagnetic longitude and latitude in the simulation are set to (37.8° N, 111.6° E), the local time is set to 0, the $F_{10.7}$ index is set to 80, and the geomagnetic index (Ap) is set to 5.

The measurement noise can greatly influence the retrieval accuracy; thus, it needs to be simulated, originating from the following: shot noise, CCD noise, and atmospheric background scattering. The shot noise is calculated from the square root of the signal with a Poisson distributed random error. The CCD noise is derived from both the dark noise and the read-out

Table 1. Satellite-based simulation parameters of FPI.

Parameters	Value
Laser wavelength	632.8168 nm
Airglow	557.7 nm, 630.0 nm, and O_2 (0–0) nm
Airglow Kinds	Dayglow and nightglow
Satellite altitude	585 km
Airglow altitude	80–300 km
Vertical field-of-view	0.05°
Horizontal field-of-view	2.5°
Telescope area	44.2 cm^2
Filter width	0.5 nm
Focal length	0.38 m
Etalon gap	2 cm
Etalon reflectivity	0.76
Integration time	1 s
CCD size	13.312 mm×13.312 mm
CCD pixel	256 × 256 (binned)
CCD noise	Dark current: 0.5 e^-/s/pixel
	Read out noise: 9 e^-/s/pixel
Shot noise	Square root of signal+Poisson distribution

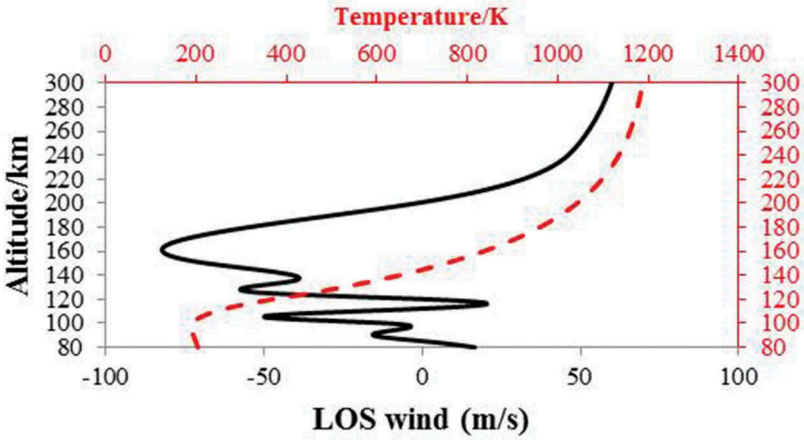

Figure 2. 80–300 km neutral wind speed (HWM07model) and temperature (IRI2012).

noise, which refer to the parameters of a commercial CCD. The atmospheric background scattering also affects the signal-to-noise and accuracy of wind retrieval. It is necessary to consider its calculation in the simulation. We computed it using the atmospheric radiation transfer model MODTRAN and the upper atmospheric density (Figure 3). The daytime background scattering mainly constitutes the atmospheric scattering of the solar radiation, while the nighttime background scattering corresponds to that of the lunar radiation. In the simulation, the atmospheric scattering of all spectra at 630.0 ± 2 nm, 557.7 ± 2 nm, and 761.1396 ± 2 nm are convoluted with a filter function. The sampling interval of these spectra is one wavenumber (corresponding to ~0.04 nm). However, Figure 3 only shows the scattering of one spectrum near 630.0 nm, 557.7 nm, and 761.1396 nm, while the scattering values of the other spectra are not shown here. Both the daytime and nighttime scattering undergo exponential attenuation with altitude, significantly decreasing from 80 km to 150 km by four to five orders of magnitude but decreasing much more slowly from 150 km to 300 km.

Once the instrument parameters, background scattering, and airglow function are known, a one-dimensional interference can be simulated. Then, the one-dimensional interference is rotated around the center of the CCD to form a two-dimensional interference ring. The CCD noise and shot noise are superimposed onto the interference ring to obtain the final simulation data. The result of the laser simulation is shown in Figure 4(a).

O_2 (0-0) dayglow simulation

The O_2 (0–0) dayglow radiation is generated from the excited state $O_2(b^1)$, which is produced through three main mechanisms (Bucholtz et al., 1986; Wallace & Hunten, 1968). (1) Atmospheric resonance scattering of solar radiation. The oxygen molecules absorb solar photons, transitioning from the ground state to the excited state. (2) Quenching reaction. The excited state $O_2(b^1)$ is obtained from the O(1D) quenching reaction with N_2 or O_2. Below the lower thermosphere, O(1D) is mainly produced from the photo-dissociation of O_2 in the Schumann region and the dissociation of O_3 in the Hartley region. (3) Barth chemical reaction. This chemical reaction includes two steps. The first is the three-body collision between two oxygen atoms and a nitrogen or oxygen molecule. The second is the generation of $O_2(b^1)$, which is derived from the quenching of the excited oxygen molecule with the ground state one.

The airglows from the different radiation mechanisms (Bucholtz et al., 1986; Wallace & Hunten, 1968) are mainly distributed at different heights (Figure 5(a)).

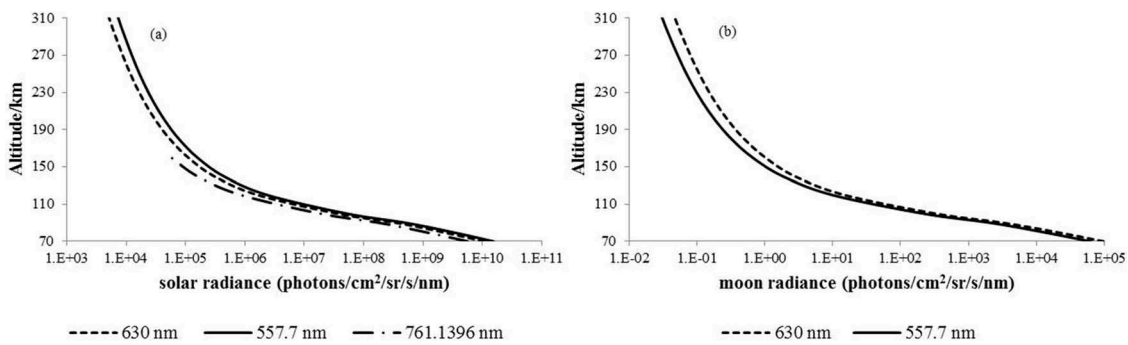

Figure 3. Atmospheric scattering from solar (a) and moon (b).

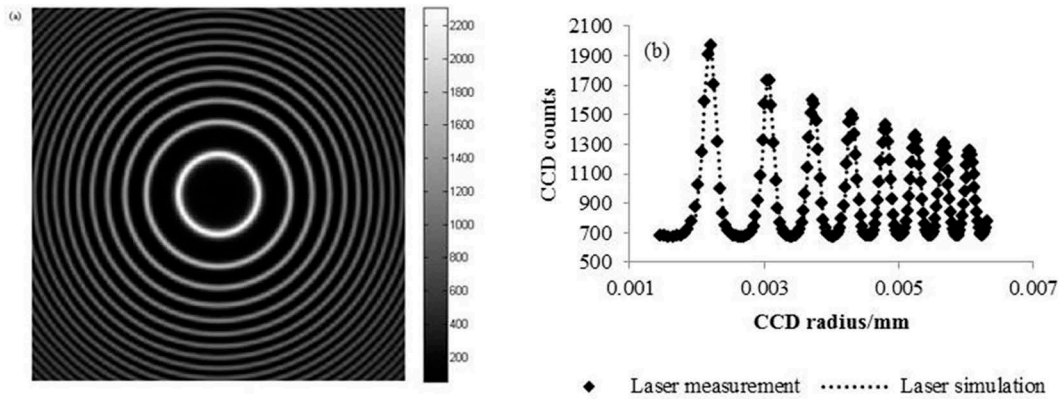

Figure 4. Simulation of laser two-dimensional interferogram (a) and curve fitting results of annular interference (b).

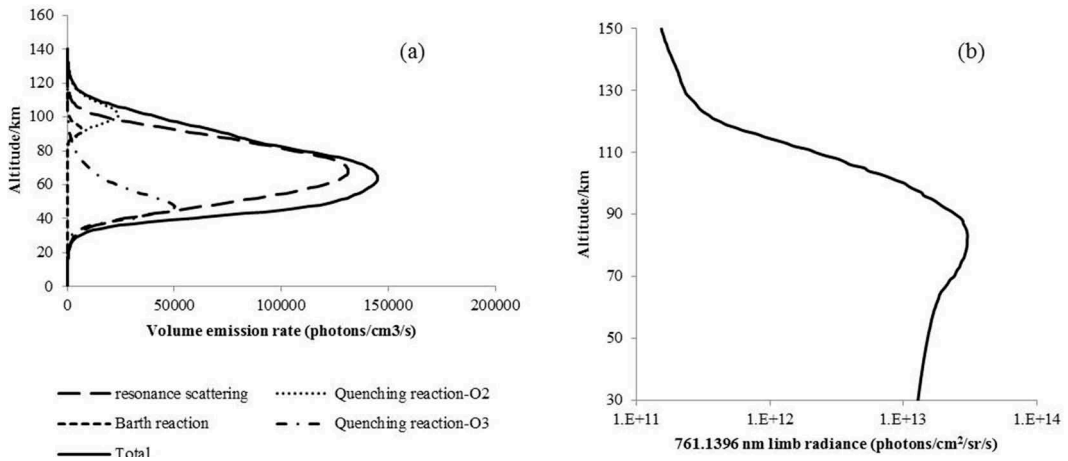

Figure 5. Volume emission rate of O_2 (0–0) dayglow (a) and 761.1396 nm limb integrated radiance (b).

The radiation from the atmospheric resonance scattering mainly occurs within 40–110 km. The radiation of the quenching reaction from O_2 photo-dissociation is primarily distributed at 80–140 km, and that from O_3 dissociation takes place at 30–90 km. The radiation from the Barth reaction mainly resides at 80–100 km. All are distributed in the range of 30–150 km. However, its nightglow becomes much weaker, and its layer thickness becomes much thinner in the range of 80–120 km, because the airglow is stimulated only by the Barth reaction.

In this study, the radiation of the O_2(0–0) dayglow is calculated by a self-constructed radiation model. Our model data are mainly based on the HITRAN database, and the methodology of the model refers to the model of Bucholtz et al. (1986). The spectral brightness B(v,z) observed near satellite is obtained by integrating the radiation along the line of sight.

$$B(v, z_t) = \int_{-\infty}^{\infty} \eta(s) f(s) D(v, s) \exp\left[-\int_{s}^{\infty} \eta(s')\sigma(s') D(v, s') ds' \right] ds \tag{7}$$

where η is volume emission rate, f is line strength, D is Doppler line shape, $n(s)$ is numerical density, and σ is cross-sectional area. According to Equation (7), the spectral integrated brightness of the O_2(0–0) dayglow is mainly determined by the volume emission rate, line strength, and Doppler line shape. Based on the equation, the brightness of all the spectral lines of the O_2 (0–0) dayglow can be obtained. However, we choose just one of the stronger spectral lines, and its central wavelength is 761.1396 nm. Its radiation is mainly distributed at the altitude of 30–150 km with the maximum at 80–90 km (Figure 5(b)).

557.7 nm and 630.0 nm airglow

The OI 557.7 nm airglow radiation is one of the strong radiation lines, produced from the forbidden transition of O (1 S→1D). Its dayglow is mainly distributed in the range of 80–300 km, while the nightglow is mainly distributed in the range of 80–120 km (dashed line in Figure 6). The OI 630.0 nm airglow is radiated by the atomic oxygen forbidden transition of O (1D→3P). Its dayglow is also a strong radiation line, which is mainly distributed in the range of 100–350 km (peak value at ~250 km), and the nightglow is mainly distributed in the range of 180–300 km (solid line in Figure 6). In this study, the volume

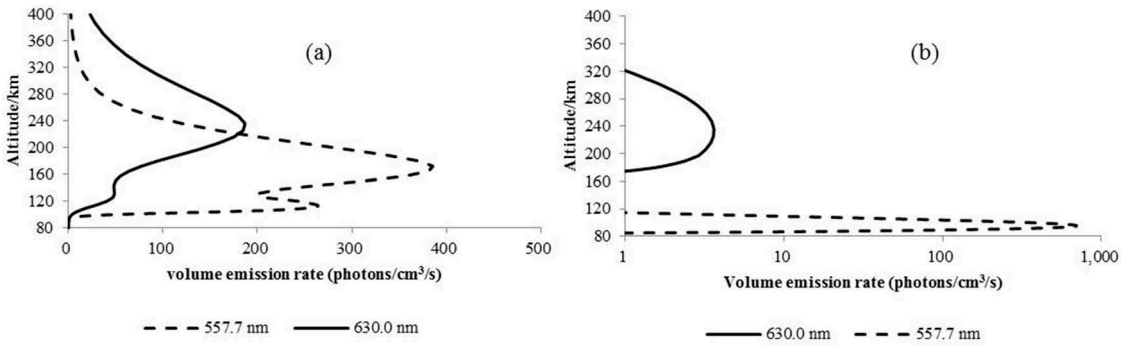

Figure 6. Volume emission rate of 557.7 nm and 630.0 nm dayglow (a) and nightglow (b).

emission rates of the 557.7 nm and 630.0 nm airglows are calculated based on the Atmospheric Ultraviolet Radiation Integrated Code model (Strickland et al., 1999; Wang & Wang, 2015).

As Figure 6 shows, the radiation of the 630.0 nm dayglow extends to above 300 km, which should be considered owing to its contribution to the lower atmosphere (≤ 300 km) radiation in actual satellite measurements. However, we assume the volume emission rate above 300 km to be zero in our simulation, because it does not affect the validation of our algorithm.

Results and analysis

For comparisons, two methods are used here for wind retrieval. One is our WDM, while the other is a traditional method proposed by Harding et al. (2014). The satellite simulated data were integrated along the line of sight, performed over a conical field-of-view. The inversion for the satellite data is based on the onion-peeling method (McDade & Lewellyn, 1993), which was widely used for limb measurements.

Owing to the strong self-absorption of the O_2 (0–0) airglow, the simulation data of < 80 km are not used to retrieve the wind here. As shown in Figure 7, the wind retrieval results from the WDM are in good agreement with the actual winds, which are input parameters in the simulation, with an average deviation of 1.41 m/s (Figure 7(a)). The traditional retrieval method has a larger average retrieval deviation of 5.16 m/s, and its randomness (or fluctuation) of errors is also larger (red dashed line, Figure 7(b)).

The wind retrieval based on the green line 557.7 nm and red line 630.0 nm airglow was also carried out. The results are shown in Figure 8, with the retrieval altitudes of the 557.7 and 630.0 nm dayglows both corresponding to 100 km to 300 km, owing to their very small volume emission rate below 100 km (Figure 6). In the range of 100–300 km, the airglow radiation from both 557.7 and 630.0 nm is strong; thus, the retrieval deviations from the WDM are both small with average deviations of 3.18 m/s and 3.29 m/s, respectively, and their average deviations from the traditional method are 3.91 m/s and 2.33 m/s, respectively. The deviations of the two methods are similar, unlike that of the O_2(0–0) airglow, because the temperature observed from the 557.7 nm and 630.0 nm airglows is due to just Doppler broadening, without the self-absorption influence.

Compared with the dayglow, the nightglow radiation of the red line 630.0 nm is much weaker, with its volume emission rate ~1/100 that of dayglow. Furthermore, the observable range of the nightglow is reduced greatly at the lowest height of ~170 km. Owing to its weak radiation, the uncertainty of the wind speed retrieval increases (H. Wang et al., 2014) (Figure 9(a)). (1) The fluctuation of the red line retrieval profile is larger, i.e., the retrieval

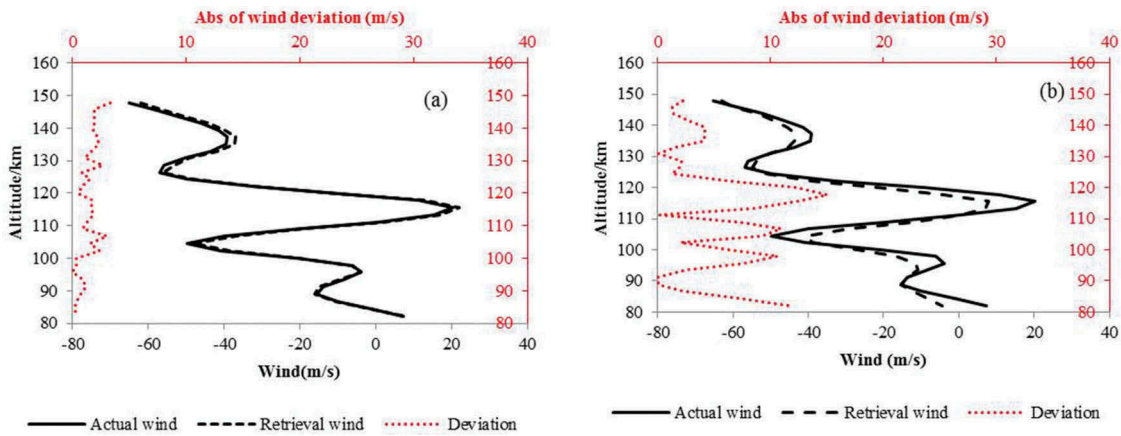

Figure 7. Wind retrieval results of 761.1396 nm using the WDM (a) and the traditional method (b).

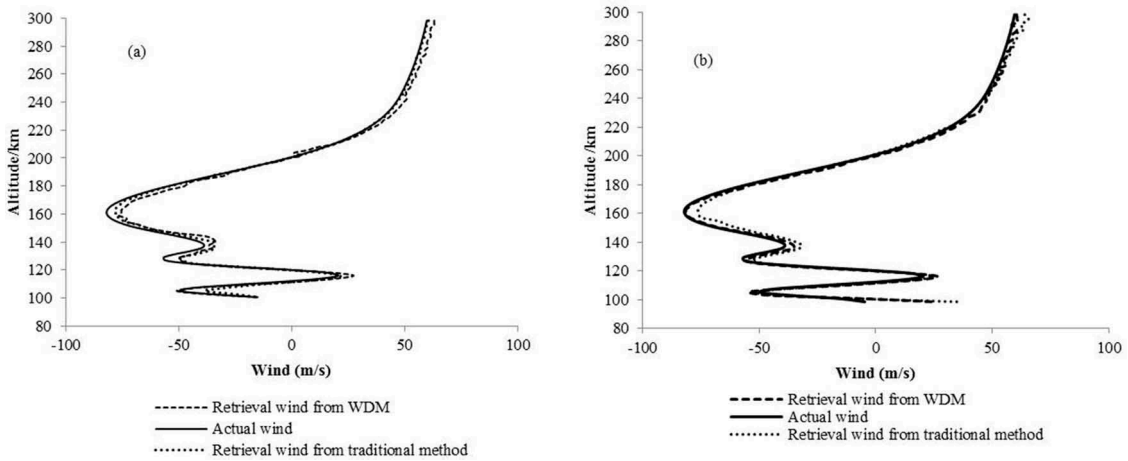

Figure 8. Wind retrieval results of 630.0 nm (a) and 557.7 nm (b) dayglow.

randomness increases. (2) The retrieval deviation is larger. The average deviation of WDM is 9.08 m/s, while the average deviation of the traditional method is 12.75 m/s. (3) In addition, the deviations at 200–250 km are a little smaller than those at other heights owing to their slightly stronger airglow. Therefore, when the airglow is weak, the retrieval uncertainty and deviations both become larger.

Unlike the red line nightglow, the radiation of the green line 557.7 nm nightglow is the same strength as its dayglow, though the distributed height becomes much smaller within the range of 80–120 km. The average deviation from the WDM is 3.79 m/s, while that obtained from the traditional method is 5.88 m/s (Figure 9(b)). The larger deviations of the traditional method mainly correspond to the top and bottom of the airglow layer whose radiation is weak, as with the 630.0 nm nightglow.

A case of ground-based FPI retrieval

For further validation, wind retrieval was carried out for the ground-based FPI measurements located in Kelan (38.71° N, 111.58° E), China (Yu et al., 2014). The FPI includes three airglows: 892.0 nm, 630.0 nm, and 557.7 nm. We obtained 15 days of data from October 12–26, 2014. The retrieval results from the WDM are compared with those obtained from the traditional method (Killeen & Hays, 1984).

The average deviations of the three airglows are 5.81 m/s, 6.03 m/s, and 1.41 m/s, respectively (Figure 10). This shows that the consistency of the 557.7 nm airglow results are the best, while the 892.0 nm and 630.0 nm airglow results demonstrate slightly larger deviations. This is mainly because of the intensity of their nightglow, rather than the retrieval method (Wang & Wang, 2015). The 557.7 nm nightglow radiation is generally stronger during the entire night. Though the 892.0 nm airglow is also stronger, its system transmittance and detector response efficiency are lower. Thus, its observed radiation is much lower. The 630.0 nm nightglow intensity varies greatly during the night in Kelan. Generally, it is very strong at dawn or dusk but becomes weakest at midnight of local time.

Discussion

In this study, based on the satellite-based simulation of the FPI, wind retrievals are carried out using the proposed method, and the results are compared with

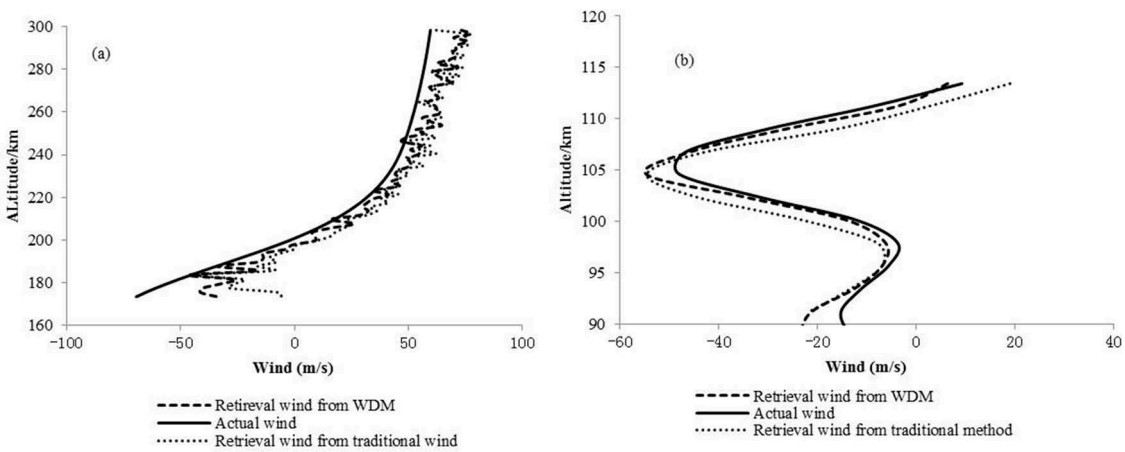

Figure 9. Wind retrieval results of redline 630.0 nm nightglow (a) and the greenline 557.7 nm (b).

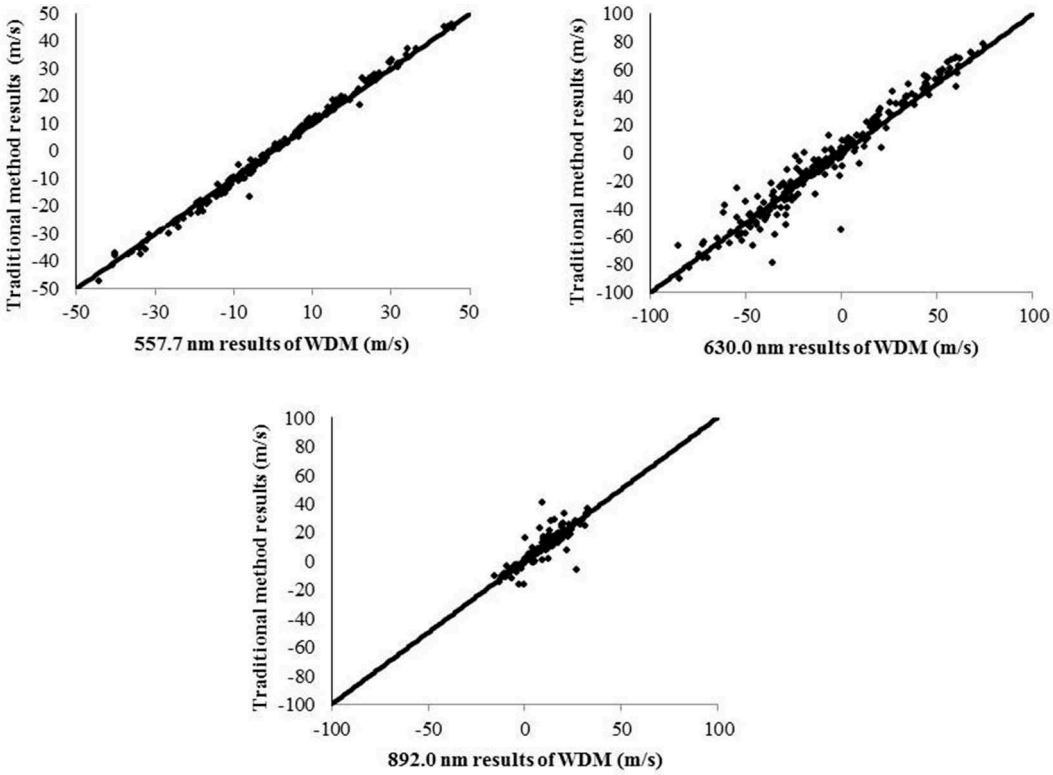

Figure 10. The comparison of results from the wavelength depth method and the traditional method.

Table 2. Wind retrieval results of satellite- and ground-based FPI.

	Time	Wavelength (nm)	Wind deviation of wavelength depth method(m/s)	Wind deviation of traditional method (m/s)
Satellite-based	Dayglow	761.1396	1.41	5.16
		557.7	3.18	3.91
		630.0	3.29	2.33
	Nightglow	557.7	3.79	5.88
		630.0	9.08	12.75
Ground-based	Nightglow	557.7	1.41	–
		630.0	6.03	–
		892.0	5.81	–

the actual values. Furthermore, the ground-based FPI measurements are also used for wind retrieval. They are summarized and shown in Table 2.

According to Table 2, we can conclude that the WDM is suitable for the wind retrieval of the self-absorption airglow. In addition, it is also applicable without the influence of self-absorption, especially when the radiation is weak. In our future work, this method will be used for the wind retrieval of other airglows with weak self-absorption at lower heights, such as that of $O_2(a^1\Delta_g)$ whose radiation model we have not yet developed.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by the National Natural Science Foundation of China under Grant [41704178].

References

Abreu, V. J., Hays, P. B., & Skinner, W. R. (1991). The high resolution doppler imager. *Optics and Photonics News*, *2* (10), 28–30. https://doi.org/10.1364/OPN.2.10.000028

Armstrong, E. B. (1969). Doppler shifts in the wavelength of the OI 6300 line in the night airglow. *Planetary and Space Science*, *17*(5), 957. https://doi.org/10.1016/0032-0633(69)90101-9

Bilitza, D. (2001). International reference ionosphere 2000. *Radio Science*, *36*(2), 261–275. https://doi.org/10.1029/2000RS002432

Bucholtz, A., Skinner, W. R., Abreu, V. J., & Hays, P. B. (1986). The dayglow of the O_2 atmospheric band system. *Planetary & Space Science*, *34*(11), 1031–1035. https://doi.org/10.1016/0032-0633(86)90013-9

Burrage, M. D., Skinner, W. R., Marshall, A. R., Hays, P. B., Lieberman, R. S., Franke, S. J., & Schmidli, F. J. (1994). Validation of winds from the high resolution doppler imager on UARS. *SPIE*, *2266*, 294–306.

Cai, M. H., Zhang, Z. L., Feng, G. Q., Zhu, L. P., & Han J. W. (2007). Study of near space neutron environment and its effects to electronic device. *Equipment Environmental Engineering*, *4*(5), 23–29.

Chen, F. G., Chen, G. M., & Liu, K. H. (2013). Analysis of near space environment and its effect. *Equipment Environmental Engineering*, *10*(4), 71–75. http://en.cnki.com.cn/Article_en/CJFDTotal-JSCX201304016.htm

Gerrard, A., & Meriwether, J. (2011). Initial daytime and nighttime SOFDI observations of thermospheric winds from Fabry-Perot Doppler shift measurements of the 630 nm OI line-shape profile. *Annales Geophysicae*, *29*(9), 1529–1536. https://doi.org/10.5194/angeo-29-1529-2011

Harding, B. J., Gehrels, T. W., & Makela, J. J. (2014). Nonlinear regression method for estimating neutral wind and temperature from Fabry-Perot interferometer data. *Applied Optics*, *53*(4), 666–673. https://doi.org/10.1364/AO.53.000666

Hays, P. B., Abreu, B. J., Dobbs, M. E., Gell, D. A., Grassl, H. J., & Skinner, W. R. (1993). The high-resolution doppler imager on the upper atmosphere research satellite. *Journal of Geophysical Research*, *98* (D6), 10713–10723. https://doi.org/10.1029/93JD00409

Hays, P. B., & Roble, R. G. (1971). A technique for recovering doppler line profiles from fabry-perot interferometer fringes of very low intensity. *Applied Optics*, *10*(1), 193–200. https://doi.org/10.1364/AO.10.000193

Hedin, A. E., Fleming, E. L., Manson, A. H., Schmidlin, F. J., Avery, S. K., Clark, R. R., & Vincent, R. A. (1996). Empirical wind model for the upper, middle and lower atmosphere. *Journal of Atmospheric and Terrestrial Physics*, *58*(13), 1421–1447. https://doi.org/10.1016/0021-9169(95)00122-0

Hedin, A. E., Spencer, N. W., & Killeen, T. L. (1988). Empirical global model of upper thermosphere winds based on atmosphere and dynamics explorer satellite data. *Journal of Geophysical Research*, *93*(A9), 9959–9978. https://doi.org/10.1029/JA093iA09p09959

Henderson, G. (1986). Fabry–perot interferometers: Cambridge Studies in modern optics 3, Cambridge University Press. *Optics & Laser Technology*, *18*(5), 274. https://doi.org/10.1016/0030-3992(86)90094-0

Hernandez, G. (1966). Analytical description of a Fabry–Perot photoelectric spectrometer. *Applied Optics*, *5*(11), 1745. https://doi.org/10.1364/AO.5.001745

Hernandez, G. (1979). Analytical description of a Fabry-Perot spectrometer 5: Optimization for minimum uncertainties in the determination of doppler widths and shifts. *Applied Optics*, *18*(22), 3826–3834. https://doi.org/10.1364/AO.18.003826

Hernandez, G., & Roble, R. G. (1995). Thermospheric nighttime neutral temperature and winds over Fritz Peak Observatory: Observed and calculated solar cycle variation. *Journal of Geophysical Research*, *100*(A8), 14647–14659. https://doi.org/10.1029/95JA00565

Killeen, T. L., & Hays, P. B. (1984). Doppler line profile analysis for a multichannel Fabry-Perot interferometer. *Applied Optics*, *23*(4), 612–620. https://doi.org/10.1364/AO.23.000612

Killeen, T. L., Wu, Q., Solomon, S. C., Orland, D. A., Skinner, W. R., Niciejewski, R. J., & Gell, D. A. (2006). TIMED doppler interferometer: Overview and recent results. *Journal of Geophysical Research*, *111*(A10), A10S01. https://doi.org/10.1029/2005JA011484

Killeen, T. L., Kennedy, B. C., & Hays, P. B. (1983). Image plane detector for the dynamics explorer Fabry-Perot interferometer. *Applied Optics*, *22*(22), 3503–3513. https://doi.org/10.1364/AO.22.003503

Kosch, M. J., Anderson, C., Makarevich, R. A., Carter, B. A., Fiori, R. A. D., Conde, M., & Davies, T. (2010). First E region observations of mesoscale neutral wind interaction with auroral arcs. *Journal of Geophysical Research: Space Physics*, *115*(A2), A02303. https://doi.org/10.1029/2009JA014697

Larsen, M. F., & Meriwether, J. W. (2012). Vertical winds in the thermosphere. *Journal of Geophysical Research: Space Physics*, *117*(A9), A09319. https://doi.org/10.1029/2012JA017843

Makela, J. J., Meriwether, J. W., Lima, J. P., Miller, E. S., & Armstrong, S. J. (2009). The remote equatorial nighttime observatory of ionospheric regions project and the international heliospherical year. *Earth, Moon, And Planets*, *104*(1–4), 211–226. https://doi.org/10.1007/s11038-008-9289-0

McDade, I., & Lewellyn, E. (1993). Satellite airglow limb tomography: Methods for recovering structured emission rates in the mesospheric airglow layer. *Canadian Journal of Physics*, *71*(11–12), 552–563. https://doi.org/10.1139/p93-084

Peterson, V. L., Zandt, T. E., & Norton, R. B. (1966). F-region nightglow emissions of atomic oxygen: 1. *Journal of Geophysical Research*, *71*(9), 2255. https://doi.org/10.1029/JZ071i009p02255

Shiokawa, K., Kadota, T., Ejiri, M. K., Otsuka, Y., Katoh, Y., Satoh, M., & Ogawa, T. (2001). Three-channel imaging fabry-perot interferometer for measurement of mid-latitude airglow. *Applied Optics*, *40*(24), 4286–4296. https://doi.org/10.1364/AO.40.004286

Shiokawa, K., Kadota, T., Otsuka, Y., Ogawa, T., Nakamura, T., & Fukao, S. (2003). A two-channel Fabry-Perot interferometer with thermoelectric-cooled CCD detectors for neutral wind measurement in the upper atmosphere. *Earth, Planets and Space*, *55*(5), 271–275. https://doi.org/10.1186/BF03351759

Shiokawa, K., Otsuka, Y., Oyama, S., Nozawa, S., Satoh, M., Katoh, Y., & Yamamoto, Y. (2012). Development of low-cost sky-scanning Fabry-Perot interferometer for airglow and auroral stydies. *Earth, Planets and Space*, *64* (11), 1033–1046. https://doi.org/10.5047/eps.2012.05.004

Skinner, W. R., Hays, P. B., & Abreu, V. J. (1987). High resolution doppler imager. *Ann Arbor: International Geoscience and Remote Sensing Symposium*, 1 (A87-53101 24-43), 673–676.

Strickland, D. J., Bishop, J., Evans, J. S., Majeed, T., Shen, P. M., Cox, R. J., & Huffmanb, R. E. (1999). Atmospheric ultraviolet radiance integrated code (AURIC): Theory, software architecture, inputs, and selected results. *Journal of Quantitative Spectroscopy and Radiative*, *62*(6), 689–742. https://doi.org/10.1016/S0022-4073(98)00098-3

Wallace, L., & Hunten, D. M. (1968). Dayglow of the oxygen A band. *Journal of Geophysical Research*, *73*(15), 4813–4834. https://doi.org/10.1029/JA073i015p04813

Wang, H., Wang, C., Wang, Y., Zhang, X., Huang, C., & Liang, S. (2017). Retrieval of middle and upper atmospheric wind based on non-full circular fringe recorded by Fabry-Perot Inteferometer. *Science China Earth Sciences*, *60*(9), 1732–1738. https://doi.org/10.1007/s11430-016-9048-1

Wang, H., & Wang, Y. (2015). Error calculation and analysis for an improved wind retrieval method based on the ground-based Fabry–Perot interferometer measurements. *Advances in Space Research*, *56*(9), 1815–1821. https://doi.org/10.1016/j.asr.2015.03.010

Wang, H., Wang, Y., & Fu, J. (2016). A tiny Fabry-Perot interferometer with postpositional filter for measurement of the thermospheric wind. *Acta Geophysica*, *64*(6), 2748–2760. https://doi.org/10.1515/acgeo-2016-0089

Wang, H., Wang, Y., & Wang, Y. (2013). Data processing of the middle and upper atmospheric wind field retrieval based on the Fabry-Perot Interferometer. *Chinese Journal of Geophysics*, *56*(4), 1095–1101. https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/cjg2.20013

Wang, H., Wang, Y., Wang, Y., & Li, Y. (2014). Wind retrieval and error analysis of the ground-based Fabry-Perot Interferometer for the middle and upper atmosphere. *Chinese Journal of Space Science*, *34*(4), 415–425. https://doi.org/10.11728/cjss2014.04.415

Wu, Q., Gablehouse, R. D., Solomon, S. C., Killeen, T. L., & She, C. (2004). A new Fabry-Perot interferometer for upper atmosphere research. *Proceedings of SPIE*, *5660*, 218–227. https://doi.org/10.1117/12.573084

Wu, Q., Wang, W., Roble, R. G., Häggström, I., & Strømme, A. (2012). First daytime thermospheric wind observation from a balloon-borne Fabry-Perot interferometer over Kiruna (68N). *Geophysical Research Letters*, *39*(14), L14104. https://doi.org/10.1029/2012GL052533

Yu, T., Huang, C., Zhao, G., Mao, T., Wang, Y., Zeng, Z., & Xia, C. (2014). A preliminary study of thermosphere and mesosphere wind observed by Fabry-Perot over Kelan, China. *Journal of Geophysical Research: Space Physics*, *119*(6), 4981–4997. https://doi.org/10.1002/2013JA019492